

Resolution Geometry on Chaotic and Dirty Experiments

Declared Regularity, Noiseless Chaos at the Exact Layer, Typed Certificates, Anchor Budgets, and Ergodic Transfer of the Metre

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September 2026

Abstract

The theorem ladder [1] conditions its differential and metric layers on regularity (H-reg), a metrological anchor (H-SI) and a resolved medium (H-med). Two objections follow: real and chaotic data are not regular, and the anchor hypotheses shift the engineering problem to the user. This paper tests both objections, including on noiseless chaos.

Two parts of the critique are confirmed. First, the *infinite-horizon* noiseless law of a chaotic family is discontinuous in its parameter, so no smooth parameter geometry exists in that limit. Second, multiplicative count loss reopens the unit fibre that a counted channel is supposed to remove.

Everything else is refuted or bounded. Noiseless chaos is expressible without any added noise, in three ways:

- the growth rate of the number of kernel classes of the n -step history observation is the topological entropy (exactly 2^n classes, $h_{\text{top}} = \ln 2$, at $r = 4$);
- the noiseless history law is a statistical experiment whose entropy locks to the Lyapunov exponent by Pesin's formula;
- every finite-horizon noiseless experiment is stratified-regular, with finitely many kneading walls and finite Fisher–Rao length across them.

An independent four-path attack is audited. Its edge-of-chaos certificate SYMBOLIC-PERIODIC is false: ordinal patterns of order three and four are exactly blind (total variation 0 against a period-four orbit), exact return fails, and the return-distance cascade exhibits Feigenbaum scaling (2.50) and a 2^7 -band chaotic attractor. A fixed-length Lyapunov confidence interval also falsely certifies chaos at the Feigenbaum point, where the exponent is zero. Typed certificates with Ruelle, variational, exact-return and length-escalation locks remove both errors.

On a dirty chaotic ranging slice, 19 of 30 windows are certified in metres (maximum error 0.68%), against a naive median error of 6.5%. Across an anchor outage, a stationary chaotic source carries the metre with a 1.0% error floor. Fifteen registered checks pass on five seeds, and eight mutation canaries are caught.

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1 The critique

C1 (regularity). Real data are rarely regular or dominated, so Sections 5–11 of the ladder do not apply, and a chaotic, dirty experiment will trip on a wall.

C2 (anchor burden). Trusting that a counted channel removes dilations (H-SI) and that the medium is resolved (H-med) shifts an unquantified burden to the user.

Claim	Verdict	Section
Infinite-horizon noiseless chaos has no smooth parameter geometry	confirmed	2
Therefore RG cannot be applied to chaotic systems	refuted	2, 3
Noiseless chaos is outside RG (“the noiseless limit is a wall”)	refuted	3
The kernel layer cannot distinguish periodic from chaotic	refuted	3
One chaos flag suffices	refuted	4
Dirty data yield no output	refuted	6, 11
Calibration effort is unquantified	refuted	7
H-med is an act of faith	refuted	8
A counted channel always removes dilation	confirmed (gap)	9
The unit is lost during an anchor outage	bounded	10

Table 1: Verdicts. “Bounded” means crossed under a testable hypothesis with a measured error floor.

2 Regularity is declared

No-Go 2.1 (Noiseless chaos). For the logistic family $x \mapsto rx(1-x)$, the law map $r \mapsto \lim_T \frac{1}{T} \sum_t \delta_{x_t}$ is not continuous: across $\Delta r = 10^{-3}$ at the period-three window ($3.828 \rightarrow 3.829$), the total-variation distance between long-run histograms is 0.568 and the Lyapunov exponent changes from 0.301 to -0.171 . Hyperbolic parameters are dense [12, 17], and linear response fails for this family [10]. For $x_0 \sim U(0,1)$ the laws of (x_0, x_1) lie on the disjoint curves $x_1 = rx_0(1-x_0)$, so distinct parameters give mutually singular laws. The experiment is neither regular nor dominated.

The exact layer of the ladder (L1–L11) needs no regularity and applies unchanged: the quotient is the set of distinct laws, and here every parameter is its own class.

Imported Theorem 2.2 (Regularity of noisy finite-horizon experiments; [22, 14]). *Let the declared record be $y_{0:T}$ with $y_t = m_t(\theta) + \varepsilon_t$, where $T < \infty$, the maps $\theta \mapsto m_t(\theta)$ are C^1 , and ε_t are independent with a smooth positive density of finite Fisher information. Then the experiment is dominated and differentiable in quadratic mean. The same holds for Markov records with Gaussian dynamical noise of fixed variance.*

Proposition 2.3 (Declared noise regularizes; MC02). *With dynamical noise $\sigma = 10^{-2}$, the total-variation distance between the two parameters of No-Go 2.1 is 0.011, and the per-step Fisher information $\mathbb{E}[(x(1-x))^2]/\sigma^2$ varies over $r \in [3.80, 3.86]$ with maximal relative jump 0.94%. It scales as σ^{-2} (4, 400, 4×10^4 for $\sigma = 10^{-1}, 10^{-2}, 10^{-3}$), which locates the non-dominated wall at $\sigma = 0$.*

Remark 2.4. The geometry obtained is that of the declared experiment. It depends on σ and T , as the ladder requires (L32). What is refuted is the inference from “the latent dynamics are chaotic” to “the experiment is irregular”.

3 Noiseless chaos at the exact and law layers

No-Go 2.1 concerns the infinite-horizon limit and the parameter geometry. It does not say that a noiseless deterministic system is invisible to RG. Declare the history protocol

$$F_n : X \rightarrow \{0,1\}^n, \quad F_n(x) = (s(x), s(fx), \dots, s(f^{n-1}x)), \quad s(x) = \mathbf{1}[x \geq \tfrac{1}{2}],$$

for a unimodal map f with critical point $\frac{1}{2}$. Time translation acts on records by the shift, so the dynamics is part of the stabilizer of the joint law.

3.1 Finite-history experiments are stratified-regular

Proposition 3.1 (Kneading walls; MC09). *Let $x_0 \sim U(0,1)$. For each n the word law $P_{r,n}$ of F_n is continuous in r and smooth off a finite set of kneading walls. These are the parameters where the critical orbit meets a cylinder boundary: finitely many roots of polynomial equations $f_r^k(\frac{1}{2}) = \frac{1}{2}$ with $k \leq n$ [7]. In $r \in [3.60, 3.99]$ there are 1, 4, 15, 46 walls for $n = 4, 6, 8, 10$. The Fisher information is finite at every sampled non-wall parameter. The Hellinger distance scales linearly with the step ($3.9 \times 10^{-3}, 4.0 \times 10^{-4}, 4.0 \times 10^{-5}$). The Fisher–Rao chord length across all walls converges (3.9467, 3.9497, 3.9505 under mesh refinement).*

Proof sketch. Cylinder boundaries are preimages of $\frac{1}{2}$ under f_r^k , $k < n$. They are continuous in r , and the set of admissible words changes only when a boundary point is created or annihilated at the critical value, i.e. on the stated finite set. Between walls, cylinder lengths are smooth in r . At a wall a new cylinder is born with length 0, so the law is continuous. $\square$

The only true noiseless wall is therefore the limit $n \rightarrow \infty$, where the walls become dense (hyperbolic parameters are dense [12, 17]). This sharpens No-Go 2.1. RG applies at every finite horizon, with a stratification that refines with n (L15 of the ladder).

3.2 Topological entropy from the kernel alone

Theorem 3.2 (Exact-layer entropy; MC10). *For a unimodal f and the protocols F_n , the kernel classes of F_n are the laps of f^n (up to at most as many boundary classes as laps). Hence*

$$h_{\text{top}}(f) = \lim_{n \rightarrow \infty} \frac{1}{n} \log |X/\mathcal{K}_{F_n}| = \inf_n \frac{1}{n} \log \ell(f^n),$$

and $\frac{1}{n} \log \ell(f^n)$ is a rigorous upper bound for every n .

Proof. F_n is constant between consecutive points of $\bigcup_{k < n} f^{-k}(\frac{1}{2})$. For unimodal maps cylinders are intervals, so distinct intervals have distinct words. These points are the turning points of f^n , so the classes are laps. The limit is the Misiurewicz–Szlenk formula [6], and the infimum follows from submultiplicativity $\ell(f^{m+n}) \leq \ell(f^m)\ell(f^n)$. $\square$

r	certificate	growth rate	comparison
4.0	positive	$\ln 2$ exactly	$ X/\mathcal{K}_{F_n} $ counts 2^n for $n < 20$
3.9	positive	0.5403	$\log \varphi \leq h \leq 0.5822$ (bound at $n = 20$), $h \geq \lambda = 0.4960$
3.835	positive	0.4823	$\log \varphi = 0.4812$ (period three [3])
3.5	upper bound	—	$h \leq 0.357$
3.56995	upper bound	—	$h \leq 0.377$

Lemma 3.3 (No zero certificate from counts). *Finite class counts never certify $h_{\text{top}} = 0$: every finite sequence of counts is compatible with arbitrarily small positive entropy. The certifier issues only POSITIVE (on a growth-rate plateau) or an upper bound. Allowing ZERO certificates certifies the edge, where $h_{\text{top}} > 0$, as zero (canary MC10).*

This refutes the claim that no statement based only on the kernel can separate periodic from chaotic dynamics. It is true for a single static observation with trivial kernel, and false for the declared history filtration, whose kernel growth is the topological entropy.

3.3 Metric entropy of the noiseless history law

The stationary history law of a noiseless system is a statistical experiment on $\{0, 1\}^n$. Its metric entropy is a natural module (a functional on law space, invariant under symbol relabeling), estimated by the conditional block entropies $h_n = H_{n+1} - H_n$, which are non-increasing and bound h_μ from above.

Imported Theorem 3.4 (Entropy relations; [5, 8, 9]). *For an ergodic invariant measure μ of a C^2 interval map, $h_\mu \leq \max(\lambda, 0)$ (Ruelle), with equality iff μ is absolutely continuous (Ledrappier). Moreover $h_\mu \leq h_{\text{top}}$ (variational principle).*

Proposition 3.5 (Law-layer locks; MC11). • $r = 4$. *The history law is iid Bernoulli($\frac{1}{2}$) ($\chi^2 = 245.6$ on 255 degrees of freedom for 8-blocks), and $h_n = 0.69312\text{--}0.69315$ against $\lambda = 0.693143 = \ln 2$.*

- $r = 3.9$. h_n decreases (0.5257, 0.5124, 0.4999, 0.4975) to within 0.0015 of $\lambda = 0.4960$ (Pecin lock).
- $r = 3.835$. $h_n < 10^{-10}$ with $\lambda = -0.310$.
- Edge, $r = 3.56995$. $h_{12} = h_{14} = 0.0580$ forms an apparent plateau, but it violates Ruelle against $\lambda = 0.00315$ by a factor 18. The law module is therefore rejected as unconverged, not certified. Removing the Ruelle lock certifies the false value (canary MC11).

Remark 3.6. When an absolutely continuous invariant measure exists, $\lambda = h_\mu$ is a function of the history law. The Lyapunov exponent is then itself law-defined and needs no declared derivative.

4 Typed certificates

Definition 4.1 (Typed certificate). A certificate $\text{Cert}_{M,W}$ names its module M and its escalation window W :

- *kernel*: class-count growth over horizons n ;
- *law*: conditional block entropy over n ;
- *derivative*: Lyapunov exponent over lengths $N, 4N, 16N$;
- *return*: exact return with tolerance at floating precision;
- *resolution*: partitions ε , or ordinal order d .

It is issued only on a plateau of its escalation, and only if it satisfies the locks: Ruelle, variational, and “PERIODIC only with exact return”.

Proposition 4.2 (Real typed disagreement; MC13). *At $r = 3.835$, inside the period-three window, the kernel module certifies topological chaos ($h_{\text{top}} = 0.4823 \approx \log \varphi$). The return module certifies an exact 3-cycle, and the derivative module certifies order ($\lambda = -0.310$). The law module gives $h_\mu < 10^{-10}$. These statements are consistent with the variational principle: topological chaos without physical chaos. One chaos flag cannot express this.*

Proposition 4.3 (Length lock at the Feigenbaum point; MC15). *At $r_\infty = 3.5699456718709\dots$, where $\lambda = 0$, fixed-length ensemble estimates are 1.5×10^{-4} , 8.2×10^{-5} and 3.3×10^{-5} for $N = 4000, 16000, 64000$. Each has a 5σ interval that excludes zero, so every single-length certificate is false. The double escalation lock ($N, 4N, 16N$ must agree) abstains.*

Just above r_∞ it certifies chaos: $\lambda = 3.58 \times 10^{-4}$ at $r = 3.5699457$ and 3.15×10^{-3} at $r = 3.56995$. The band-level exponents $\lambda 2^m$ ($2^m = 1024$ and 128 bands from band-merging scaling) are 0.366 and 0.404.

A single $N \rightarrow 4N$ lock at short N passed the Feigenbaum point on three of five seeds. It was replaced; see the appendix.

Proposition 4.4 (Sweep; MC14). *Over 30 parameters in $[3.5, 4.0]$, including r_∞ , the typed certifier issued no certificate violating Ruelle, the variational principle or exact return, and abstained at r_∞ . A naive certifier (ordinal order three plus Lyapunov sign) issued PERIODIC on 8 aperiodic parameters.*

5 Audit of the four-path attack

An independent attack [4] proposed four routes into noiseless chaos (Lyapunov module, symbolic ordinal law, finite-precision quotients, regularizing noise) and certified four windows on five seeds. Each claim, tested:

Four-path claim	Verdict	Evidence
$r = 3.56995$: CHAOS (Lyapunov)	confirmed	$\lambda = 0.00315$, length-locked (MC15)
$r = 3.56995$: SYMBOLIC-PERIODIC	false	ordinal TV = 0 at orders 3–4; no exact return up to 2^{16} ; 2^7 -band chaos (MC12)
Identical symbolic statistics at $r = 3.5$ and 3.56995	explained	order blindness, not a coincidence
$r = 4$: entropy rate 0.271	not locked	true $h = \lambda = \ln 2$; fixed-order ordinal entropy is not an entropy certificate [2]
Finite-precision ABSTAIN at the edge	confirmed	consistent with Lemma 3.3
Noise path: REGULAR-NOISY-LAW, wall at $\sigma = 0$	confirmed	duplicates Proposition 2.3
“No theorem from $\mathcal{K}$, Q , static Γ separates periodic from chaotic”	refuted	Theorem 3.2
“No single chaos flag”	confirmed, sharpened	MC13: topological versus physical
Block-bootstrap Lyapunov CI certifies CHAOS	fails at r_∞	Proposition 4.3

Proposition 5.1 (The edge is 2^7 -band chaos; MC12). *At $r = 3.56995$ the return distance $\max_t |x_{t+p} - x_t|$ contracts by 2.502, 2.496 and 2.471 for $p = 4 \rightarrow 8 \rightarrow 16 \rightarrow 32$ (Feigenbaum $\alpha = 2.5029$). It plateaus at 1.25×10^{-3} from $p = 128$. Band-merging scaling predicts $2^{[6.57]} = 128$ bands. There is no exact return for $p \leq 2^{16}$, and $\lambda = 0.00315 > 0$. The PERIODIC certificate would require exact return, which fails. Loosening the return tolerance to band-width scale (5×10^{-3}) certifies the edge as periodic (canary MC12).*

6 Walls are located, not fatal

Definition 6.1 (Certified window atlas). Partition the record into windows. A window is *certified regular* if a lower confidence bound of the smallest singular value of the estimated response exceeds a declared tolerance τ . The certified windows, with walls between them, form the atlas on which the ladder’s differential layers are applied.

Proposition 6.2 (Controlled certification; MC03). *Consider a three-channel, two-parameter response observed through Student- t_3 noise with 5% gross outliers. Use Huber regression [13] and bootstrap lower bounds [11] at level $\alpha = 0.05$ with $\tau = 0.3$. In 100 wall windows and 100 regular windows, the false-certification rate was 0.00 and the power 1.00. At 60% outliers the certification fails, as it must.*

Lemma 6.3 (Invariance is certified by equivalence, not by non-rejection). *A statement “structure X is unchanged” may be certified only by an equivalence test: an upper confidence bound of the change must lie below a declared tolerance [19]. Failure to reject a change is not certification.*

Remark 6.4. In the slice of Section 11, replacing the equivalence test of detector isotropy by non-rejection lets a stale calibration be carried into an anisotropic regime, with distance-ratio errors above 2% (canary MC07). With 200 samples per window the equivalence test certifies no carry-forward: the data cannot establish unchanged isotropy to 2%, and the procedure abstains.

7 The anchor budget

Theorem 7.1 (Anchor budget; MC04). *Let h be a constant metric on a target chart V whose empirical isotropy group is $K \leq GL(V)$. Radar anchors $\Delta_a^\top h \Delta_a = \rho_a^2$ determine h iff their number in general position is at least*

$$b(K) = \dim(\text{Sym}^2 V^*)^K.$$

With fewer anchors, a family of dimension $b(K) - k$ fits them.

Proof. The admissible metrics form the linear space $(\text{Sym}^2 V^*)^K$. Each anchor is one linear equation in h ; in general position k equations have rank $\min(k, b(K))$. $\square$

Corollary 7.2. *In two dimensions $b = 1$ for $SO(2)$ and for C_4 , and $b = 3$ for C_2 or for no symmetry. In three dimensions $b = 1$ for an irreducible rotation group and $b = 6$ for none. Computed ranks with $k = 1, \dots, 4$ anchors are $1, 1, 1, 1$ ($SO(2)$) and $1, 2, 3, 3$ (C_2). With two anchors and no symmetry, $h = I$ and $h = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}$ fit both anchors and differ by 0.6 on a held-out pair.*

The number of anchors a dirty experiment needs is therefore a computed output of its empirical symmetry, not an open-ended effort.

8 The medium as a kernel direction

Proposition 8.1 (Frequency diversity; MC05). *For two-way delay $\tau(f) = 2d/c + \sum_{p \in P} 2K_p/f^p$, the parameters (d, K_p) are identifiable iff the number of frequencies is at least $|P| + 1$, with the Vandermonde-type rank condition. With $P = \{2\}$, one frequency gives rank 1, and two give rank 2 (condition number 9.2 for 1.5 and 1.2 GHz). With $P = \{2, 3\}$, two frequencies leave an exact kernel direction, and three give rank 3 (condition number 848). An unmodelled dispersion term is an unremoved fibre.*

Proposition 8.2 (Path diversity; MC05). *For non-dispersive excess path with range $\rho_e = \rho_0 - h \sin e + Z/\sin e$ over elevations e [18], a single elevation leaves (h, Z) confounded. Over twelve elevations from $e_{\min}$ to 90° the condition number is 2322, 110, 37.7, 28.7 for $e_{\min} = 60^\circ, 30^\circ, 15^\circ, 5^\circ$. The cost of H -med is this number.*

Definition 8.3 (Integer fibre). After loss of lock, a counted carrier record carries an unknown integer offset per channel. The integer vector is a discrete nuisance fibre of the joint law.

Proposition 8.4 (Ratio-tested resolution; MC07; [20, 23]). *Nonlinear integer least squares on the accumulated trilateration misclosure, accepted only when the ratio of second-best to best cost exceeds 3, resolved every re-acquisition in the slice of Section 11 (three stations: 3 accepted, 0 wrong; four stations: 3 accepted, 0 wrong). Until acceptance the procedure abstains. Redundancy raises the ratio margin: 21.5 with four stations versus 6.0 with three at the first re-acquisition.*

9 When a counted channel does not remove dilation

No-Go 9.1 (Scale-covariant count loss; MC06). Let the counted channel record $n = kd$ with $k = 2\Delta\nu_{Cs}/c$ and loss mechanism L . If L is multiplicative, $n' = (1 - \epsilon)kd$ with unknown ϵ , then (d, ϵ) has an exact dilation fibre: Fisher rank 1 in (scale, ϵ) over any set of anchor distances. Additive slips $n' = kd - s$ and dead time $m = n/(1 + n\tau_d)$ [15] give rank 2 when the anchor distances, and hence the count rates, vary.

Hypothesis 9.2 (H-SI, refined). In addition to the clauses of the ladder, the count-loss mechanism of the anchor channel is either absent or not scale-covariant, and its parameters are identified jointly with the anchor.

This clause was missing from the ladder. Without it, Lemma L8 (count rigidity) is true, but its consequence L48 (dilation removed) can fail.

10 Crossing an outage with a chaotic source

During an anchor outage, a change of detector scale is a pure dilation and is invisible to every imaging-only natural module (L47). One further structure can carry the unit: a stationary source.

Hypothesis 10.1 (H-stat). Time translation belongs to Γ on the source channel across the outage: the source law, and hence its invariant measure μ , is the same before and after.

Theorem 10.2 (Ergodic transfer). *Assume H-stat, and that the calibrated segment gives physical positions with covariance Σ and third central moment tensor T of μ . Let $G_T = \{L : L\Sigma L^\top = \Sigma, L \cdot T = T\}$. If the outage record is $y = Mp + b + \text{noise}$, then M is determined by the second and third moments of y up to right multiplication by G_T . If $G_T \subset O(2)$, all distances $\|M^{-1}\Delta y\|$ are determined.*

Proof. $\text{Cov}(y) = M\Sigma M^\top$ gives $M = \text{Cov}(y)^{1/2}O\Sigma^{-1/2}$ with O orthogonal. The third moments give $M \cdot T = T_y$; two solutions differ by $L \in G_T$. If L is Euclidean-orthogonal, $\|L^{-1}p\| = \|p\|$. $\square$

Remark 10.3. The condition $G_T \subset O(2)$ is checkable on the calibrated segment. For the Lorenz attractor the residual symmetry is the reflection $x \mapsto -x$, an isometry.

Proposition 10.4 (Stationarity test). *Two families of invariants are unaffected by any spatial linear map and test H-stat across the outage: (i) affine-invariant standardized moments of the whitened record; (ii) the mean interval between maxima of the whitened radius, read on the clock channel, which runs during the outage. Family (ii) is the powerful one. At record length 3000 time units, a drift $\rho : 28 \rightarrow 30$ gives $z = -13.2$ on (ii) and 1.8 on (i).*

Theorem 10.5 (Transfer certificate; MC08). *A transfer is certified if neither invariant family is flagged ($|z| \leq 3$). The certificate carries an error floor ε^* : the largest distance error among unflagged drifts at the given record length. At 3000 time units, with an unknown dilation of 3% and anisotropy 1.25:*

ρ after outage	median distance error	period z	moment $\max z $	flagged
28.00	0.50%	-0.6	0.9	no
28.15	0.60%	-1.6	1.2	no
28.50	1.01%	-2.9	0.4	no
29.00	1.96%	-7.1	0.6	yes
30.00	3.76%	-13.2	1.8	yes

so $\varepsilon^* = 1.0\%$, and no drift with error at least 3% went unflagged. On seeds 1, 42 and 777, ε^* was 0.57%, 0.46% and 0.47%, and $\rho = 28.5$ was flagged. Without the clock invariant, the drift to $\rho = 30$ (error 3.8%) is not flagged (canary MC08).

Remark 10.6 (What is crossed and what is not). The unit is carried across the outage only under H-stat, which is tested but not proved by the invariants. Drift below their sensitivity biases the result by at most ε^* . That bound is a boundary of the data, not of the method, and it shrinks with record length. Without a stationary source, the outage dilation is not crossed.

11 A chaotic dirty slice

Construction (MC07).

- *Source.* A target follows the Lorenz attractor [16] projected to a 21×19 m region.
- *Radar.* Three stations (then four) range the target by two-way dual-frequency timing (1.5 and 1.2 GHz, 30 ps noise) through a turbulent dispersive medium with about 3 m of delay. Cycle slips of ± 1 –3 cycles occur at a rate of 1/400, and there are three radar outages.

- *Imaging.* Student- t_3 position jitter (2 cm scale), 3% gross outliers and 2% dropouts.
- *Detector regimes.* 6000 samples in three regimes: isotropic; anisotropic ($R(0.3) \text{diag}(1, 1.25)$); anisotropic with a 3% pixel-pitch dilation.

Procedure per window (200 samples).

1. Repair cycle slips continuously within each tracking segment.
2. After re-acquisition, resolve the integer fibre with the ratio test; abstain until it is resolved.
3. Form the ionosphere-free combination and trilaterate.
4. Certify the radar by its misclosure.
5. Calibrate the detector robustly on the first half-window; certify by held-out distance prediction on the second half (1%).
6. In outages, carry the calibration only under an equivalence test of detector isotropy (Tyler scatter [21]) (Lemma 6.3); otherwise abstain.

	3 stations	4 stations
windows certified in metres	19/30	19/30
median / max distance error, certified	0.49% / 0.68%	0.45% / 0.98%
outage windows certified in metres	0	0
wrong integer fixes accepted	0	0
naive global calibration: median / 90th percentile	6.5% / 18.2%	9.8% / 23.6%
all-or-nothing certification	reject	reject

Table 2: Window pattern (three stations): MMMMMMMM...M...MM...MMMM.MM, where M is certified in metres and “.” is abstention. Abstentions are outages and windows awaiting integer resolution. Seeds 1, 42, 777: 13–17 windows certified, maximum certified error $\leq 0.91\%$, 0 wrong fixes.

The procedure does not certify the record as a whole; it certifies an atlas. Where it abstains, the reason is named: an outage, an unresolved integer fibre, or unestablished isotropy.

12 Procedure for chaotic and dirty experiments

1. *Declare the experiment.* Include the noise model and the horizon; regularity is a property of this declaration (Imported Theorem 2.2).
2. *Window the record.* Certify regular windows with lower confidence bounds; walls are abstentions (Proposition 6.2).
3. *Test Γ per window.* Certify invariance only by equivalence tests (Lemma 6.3).
4. *Compute the anchor budget $b(K)$* from the empirical isotropy (Theorem 7.1).
5. *Remove medium fibres.* Use frequency diversity for dispersive terms and path diversity for non-dispersive ones; report condition numbers.
6. *Treat integer ambiguities as a discrete fibre.* Accept a fix only by ratio test.
7. *Check the count-loss mechanism* against refined H-SI (Hypothesis 9.2).

8. *Across outages*, transfer the unit only under **H-stat**, with clock-based and affine invariants, and report the floor ε^* (Theorem 10.5).
9. *For dynamics*, declare the history protocol. Certify with typed modules on escalation plateaus only, enforce the Ruelle, variational and exact-return locks, and never certify zero entropy from finite counts.
10. *Validate every certified window on held-out anchors*.

13 Boundaries that remain

1. **Infinite-horizon noiseless parameter geometry.** There is none (No-Go 2.1); every finite horizon is stratified-regular (Proposition 3.1).
2. **Zero entropy from finite data.** It is never certified (Lemma 3.3); near r_∞ all modules need escalation lengths of order $1/h$ and abstain before that.
3. **Scale-covariant count loss.** It reopens the unit fibre unless the loss is identified (No-Go 9.1).
4. **Outage dilation without a stationary source.** It is not crossed.
5. **Sub-threshold source drift.** It biases an ergodic transfer by up to ε^* .
6. **Unresolved integer fibres.** With poor geometry diversity, or with offsets beyond the search, the procedure abstains.
7. **Unmodelled dispersion, or non-dispersive excess with no path diversity.** These are exact kernel directions.

14 Ladder amendments

Where	Amendment	Status
after L8	count rigidity does not imply dilation removal under scale-covariant loss	no-go
H-SI	add the count-loss clause (Hypothesis 9.2)	hypothesis
after L51	anchor budget $b(K) = \dim(\text{Sym}^2 V^*)^K$	Γ -conditional
after L11	invariance certified only by equivalence tests	forced (statistical)
after L55	ergodic transfer under H-stat , with error floor	hypothesis-conditioned
after L16	finite-history noiseless experiments are stratified-regular; wall only at $n = \infty$	forced / imported
after L3	$h_{\text{top}} = \lim \frac{1}{n} \log X/\mathcal{K}_{F_n} $ for history filtrations	imported (Misiurewicz–Szlenk)
after L27	Ruelle, Pesin and variational locks between law and derivative modules	imported
after L57	typed certificates with escalation and cross-module locks	forced (statistical)

A Reproducibility

`rg_chaos_attack.py` passes 15/15 registered checks on seeds 20260930, 1, 42, 777 and 2026. `--canary` sabotages eight structures and each is caught:

- MC02: $\sigma = 0$ (infinite Fisher information);
- MC03: 60% outliers;

- MC07: equivalence test disabled;
- MC08: clock invariant removed;
- MC10: zero-entropy certificates allowed;
- MC11: Ruelle lock removed;
- MC12: band-scale return tolerance;
- MC15: length lock removed.

The four-path result files are included unchanged in `four_path_audit/`.

Statements corrected during development are recorded in the JSON:

1. Per-window slip repair left accumulated offsets; it was replaced by continuous tracking and integer resolution.
2. Linearized integer search gave near-unit ratios; it was replaced by nonlinear integer least squares.
3. The claim that three stations cannot resolve the integer fibre was an artifact of that search and was withdrawn.
4. The prediction that $\rho = 28.5$ is flagged failed at the primary seed ($z = -2.9$). The certificate was re-registered as an error-floor statement.
5. The Feigenbaum-ratio gate first included $p = 32 \rightarrow 64$, inside the crossover to the band plateau (2.36). It was re-registered for $p < \text{onset}/4$, the range where renormalization scaling is claimed.
6. A single-escalation Lyapunov lock at $N = 4000$ passed r_∞ on three of five seeds. It was replaced by the double escalation.
7. The MC10 canary first failed by a crash, not by a verdict. The MC12 canary was too mild to trip the gate. Both were fixed.

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